

The University of Nottingham
Malaysia Campus

SCHOOL OF COMPUTER SCIENCE

A LEVEL 1 MODULE, AUTUMN SEMESTER 2015-2016

MATHEMATICS FOR COMPUTER SCIENTISTS

Time allowed ONE Hour

Candidates must NOT start writing their answers until told to do so

There are THREE questions in the exam paper. Answer all THREE questions.

Only silent, self-contained calculators with a single-line display are permitted in this examination.

Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

DO NOT turn examination paper over until instructed to do so

1. Questions on Logic, Sets, Functions, Matrices, and Relations.

(a) Let A be the set $\{1,2,3\}$. The following relations are subsets of $A \times A$,

$$R_1 = \{(x, y) \mid x \in A \wedge y \in A \wedge (x - y = 1)\},$$

$$R_2 = \{(x, y) \mid x \in A \wedge y \in A \wedge (x - y \geq 1)\},$$

$$R_3 = \{(x, y) \mid x \in A \wedge y \in A \wedge \neg(x - y = 0)\},$$

$$R_4 = \{(x, y) \mid x \in A \wedge y \in A \wedge \forall u \exists v (x + u = y + v)\} \text{ where } u \text{ and } v \text{ are integers, and}$$

$$R_5 = \{(x, y) \mid x \in A \wedge y \in A \wedge \exists u \forall v (x + u = y + v)\} \text{ where } u \text{ and } v \text{ are integers.}$$

(i) List all the elements in R_1 and R_2 .

[4 marks]

(ii) Draw a Venn diagram to show the relations among R_1 , R_2 and $A \times A$.

[2 marks]

(iii) Represent relation R_3 using Boolean matrix. Explain whether it is symmetric.

[4 marks]

(iv) Are sets R_4 and R_5 finite? Is it true that $R_4 = R_5$? Show your work.

[6 marks]

(b) Let f and g be functions from $A = \{1, 2, 3, 4\}$ to $B = \{a, b, c, d\}$ and from B to A , respectively, with $f(1) = d$, $f(2) = c$, $f(3) = a$, and $f(4) = b$, and $g(a) = 2$, $g(b) = 1$, $g(c) = 3$, and $g(d) = 2$.

(i) Is f one-to-one? Is g one-to-one?

[4 marks]

(ii) Is f onto? Is g onto?

[4 marks]

(iii) Which one f and g has an inverse function, and what is this inverse?

[4 marks]

(iv) Let $S_f: P(A) \rightarrow P(B)$, where $S_f(X) = \{y \mid \forall x \in X, y = f(x)\}$. $P(A)$ and $P(B)$ are power sets. Determine the values of $S_f(\{1, 2\})$ and $S_f(\emptyset)$.

[5 marks]

2. Questions on Numbers, Sequences, Matrices, and Induction.

(a) Assume a, b, m, n are integers, and $a \equiv b \pmod{m}$ (i.e., a is congruent to b modulo m).

(i) Show that if $n \mid m$, then $a \equiv b \pmod{n}$.

[6 marks]

(ii) Let $a = 3^7 \cdot 5^3 \cdot 7^3$, $b = 2^{11} \cdot 3^5 \cdot 5^9$. Find the greatest common divisor and the least common multiple of a and b respectively.

[6 marks]

(b) List the first 4 terms of each of the following sequences.

(i) The n -th term is given by $\lfloor n/2 \rfloor + \lceil n/2 \rceil$.

[6 marks]

(ii) The sequence whose first two terms are 1 and 5 and each subsequent term is the sum of the two previous terms.

[6 marks]

(c) Suppose that $A = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$, where a and b are real numbers. Show that $A^n = \begin{bmatrix} a^n & 0 \\ 0 & b^n \end{bmatrix}$ for every positive integer n , using mathematical induction.

[9 marks]

3. Questions on Sets, Functions, Counting, and Probability.

(a) Let the set $A = \{1, 2, \dots, 5\}$ and the set $B = \{0, 1\}$. Let $f: A \rightarrow B$ denote a total function from A to B , i.e., every element in A is mapped to an element in B .

(i) How many possible functions f are there from A to B ?

[6 marks]

(ii) Among the results in (i), how many functions are onto?

[6 marks]

(iii) Among the results in (i), how many functions are there that assign 0 to both 1 and 5?

[6 marks]

(b) A pair of dice is rolled. The probability that a 4 appears on the first dice is $2/7$, and the probability that a 3 appears on the second dice is $2/7$. Each of the other outcomes has probability $1/7$. What is the probability of 7 appearing as the sum of the numbers on the two dice?

[8 marks]

(c) Show that if seven integers are selected from $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, there must be at least two pairs of these integers with the sum 11.

[8 marks]